

Day 5: Network Testing Tools Theory (30 min):Tools for network testing: Ping, Traceroute, nslookup.**Practical (30 min):**Use these tools to test connectivity in a LAN.**Task: Report on the tools used and results obtained.**

Task 1.

Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

```
C:\Users\sketan>ping www.youtube.com

Pinging www.youtube.com [142.251.150.4] with 32 bytes of data:
Reply from 142.251.150.4: bytes=32 time=53ms TTL=117
Reply from 142.251.150.4: bytes=32 time=74ms TTL=117
Reply from 142.251.150.4: bytes=32 time=42ms TTL=117
Reply from 142.251.150.4: bytes=32 time=33ms TTL=117

Ping statistics for 142.251.150.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 33ms, Maximum = 74ms, Average = 50ms
```

Average Response Time 50 ms

Task 2.

Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.

```
C:\Users\sketan>tracert www.flipkart.com

Tracing route to flipkart.com [163.53.76.86]
over a maximum of 30 hops:

  1    8 ms     1 ms     2 ms    10.110.11.199
  2   67 ms    39 ms    39 ms    106.213.225.1
  3   64 ms    50 ms    28 ms    dsl-tn-dynamic-237.222.22.125.airtelbroadband.in [125.22.222.237]
  4  102 ms    39 ms    60 ms    116.119.46.40
  5  100 ms    39 ms    82 ms    aes-static-082.27.22.125.airtel.in [125.22.27.82]
  6    *         *         *      Request timed out.
  7    *         *         *      Request timed out.
  8    *         *         *      Request timed out.
  9    *         *         *      Request timed out.
 10  115 ms    38 ms    59 ms    163.53.76.86

Trace complete.
```

Task 3.

Use the nslookup tool to find the IP address of www.zomato.com and record the result.

```
C:\Users\sketan>nslookup www.zomato.com
Server: UnKnown
Address: 10.110.11.199

Non-authoritative answer:
Name: e70929.dsca.akamaiedge.net
Addresses: 2600:140f:1e00::1737:f4c3
           2600:140f:1e00::1737:f4d0
           23.55.244.138
           23.55.244.115
           23.55.244.120
Aliases: www.zomato.com
          zomato.edgekey.net
```

Task 4.

Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

```
C:\Users\sketan>ping 10.110.11.156

Pinging 10.110.11.156 with 32 bytes of data:
Reply from 10.110.11.156: bytes=32 time<1ms TTL=128
Reply from 10.110.11.156: bytes=32 time<1ms TTL=128
Reply from 10.110.11.156: bytes=32 time<1ms TTL=128
Reply from 10.110.11.156: bytes=32 time<1ms TTL=128

Ping statistics for 10.110.11.156:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Task 5.

Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.

- The ping command is useful for checking whether a website or device is reachable and for measuring response time; in my test, www.youtube.com responded with an average time of 50 ms.
- The tracert command is useful for finding the network path and identifying where delays or failures occur; my test to www.flipkart.com showed 30 hops.
- The nslookup command is useful for diagnosing DNS problems because it converts a domain name into an IP address; my test returned 10.110.11.199 for www.zomato.com.
- For a website that is not loading, I would first use ping to check basic connectivity.
- For a slow connection, tracert can help identify a slow or problematic hop.
- If I suspect DNS problems, nslookup is the most useful tool because it directly checks DNS resolution.